

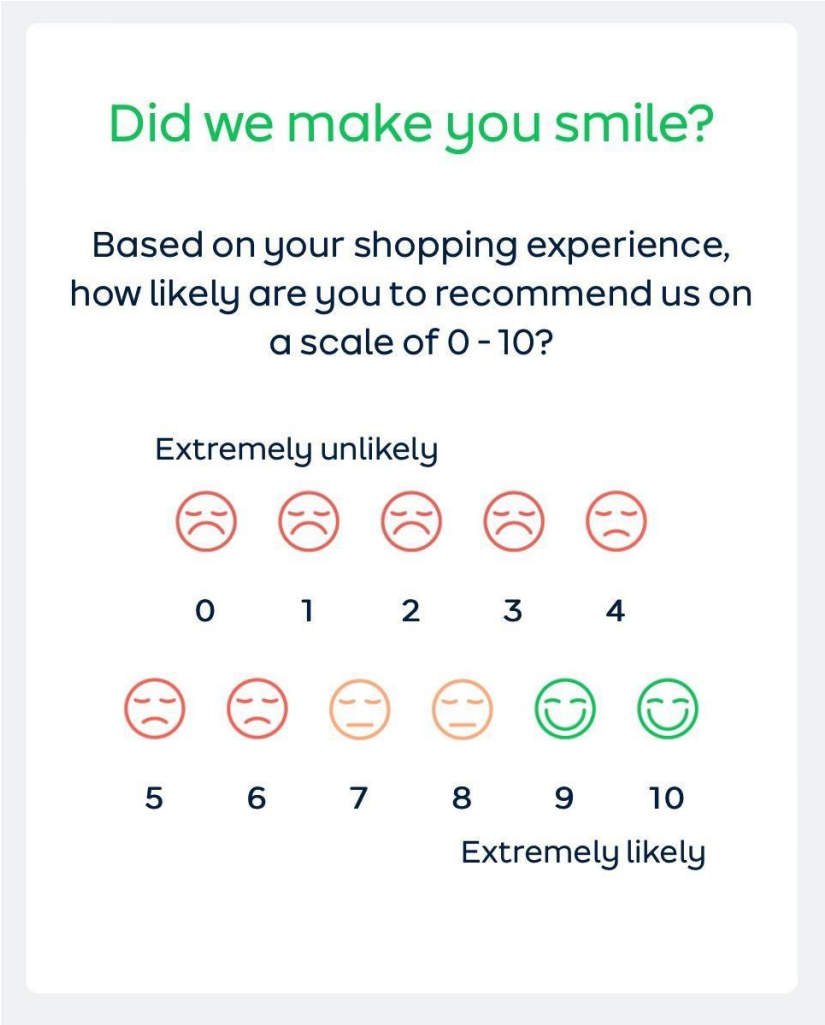
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Design Principles

Human Computer Interaction

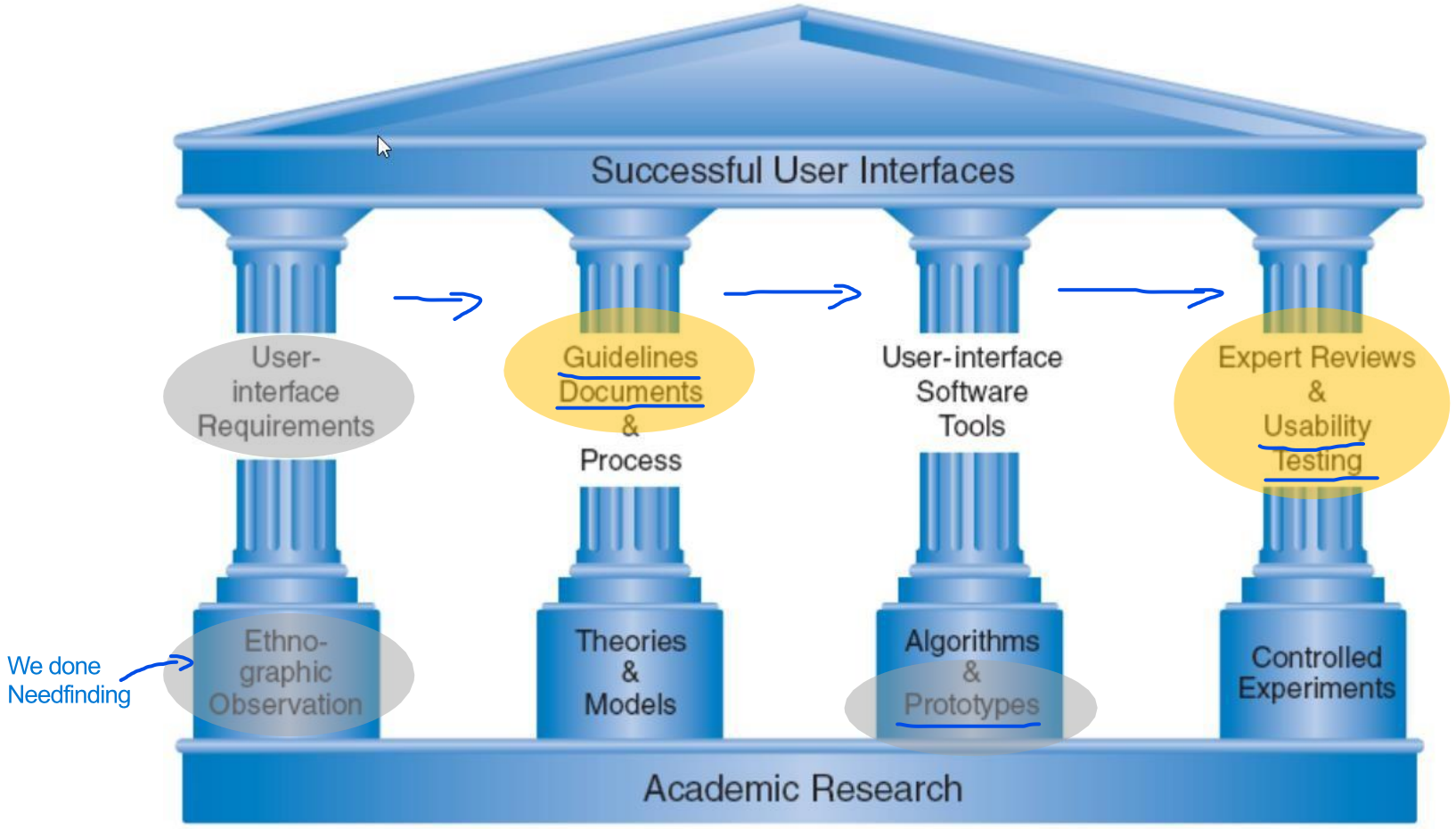
Catia Prandi
catia.prandi@unibo.it

Hall of Fame or Shame?



Nice to use emoticons and colors, but too much red ones in respect to yellow and green

The Four Pillars of Design



Ben Shneiderman & Catherine Plaisant, Designing the User Interface: Strategies for Effective Human-Computer Interaction

Goals

Design principles are not only for prototyping (generate design solutions), but even for evaluation (if you are the expert)

Generating design solutions

- Guidelines are more specific
- Principles
- Theories are more broad

Evaluating generated designs

- Expert reviews following ~~and~~ heuristics
- Usability testing
- Controlled experiments

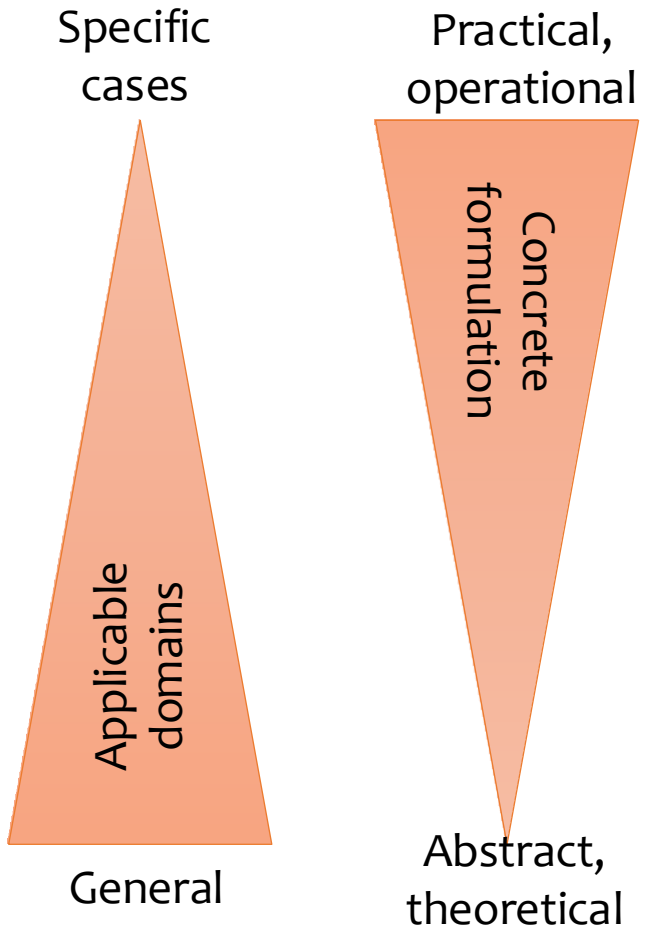
Generating Design Solutions

Very specific

- **Guidelines**: Low-level focused advice about good practices and cautions against dangers (e.g., web accessibility alternative text)
- **Principles**: Mid-level strategies or rules to analyze and compare design alternatives (not so clear: we use heuristics)

More general

- **Theories**: High-level widely applicable frameworks to draw on during design and evaluation, as well as to support communication and teaching (From theory, for a specific case, you need to define principles and then guidelines)



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Design Theories

Theoretical frameworks enabling foundational research

The "Why" e.g. why using HCI? Because we are focused on users



As we can see, Theories are high-level (not specific how to do it)

A lot of theories: someone related to human capacity someone related to what we can do with an interface considering different aspects

Design Theories

Types of theories

to understand human and human capabilities

- Descriptive
  - UI elements, terminology, semantics
    - that you gonna use
    - used in UI
- Explanatory — task: set of activities/events to achieve a goal
  - Sequences of events with causal relationships
- Prescriptive
  - Guidelines for designers to make decisions (to optimise usability to user)
- Predictive — help to predict the behaviour of the user
  - Comparison of design alternatives based on performance figures

Human capacity

(about limits and advantages of human capacity to design better)

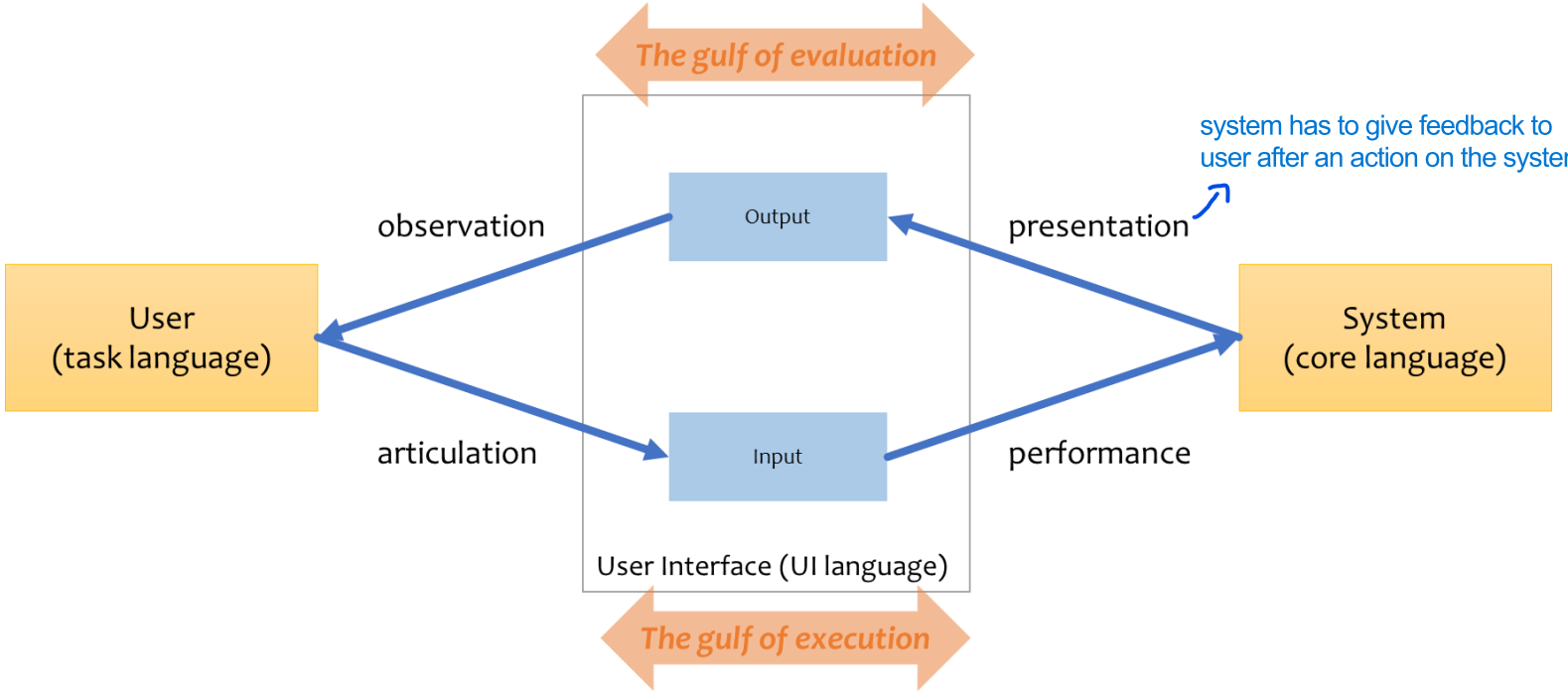
- Motor task
  - Skill in pointing, clicking, … movements
- Perceptual
  - Sensory inputs
- Cognitive
  - Problem-solving, short-/long-term memory

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Explain what happens between a human and the interface
and how these events are important to have a result

Explanatory Design Theory

Norman's Action Models (Explanatory)



Which are the steps to achieve user's goal of the task?

1. **Goal** (form the goal)
2. **Plan** (the action)
3. **Specify** (an action sequence)
4. **Perform** (the action sequence)
5. **Perceive** (the state of the world)
6. **Interpret** (the perception)
7. **Compare** (the outcome with the goal)

Consistency Theories (Prescriptive)

- **Consistency** of nouns (objects) and verbs (actions)
  - Reduces learning time and errors
- Consistency of
  - Color
  - Layout
  - Icons
  - Fonts and Font sizes
  - Button sizes
  - …

means that a specific button or semantic must be the same

Non sempre!!

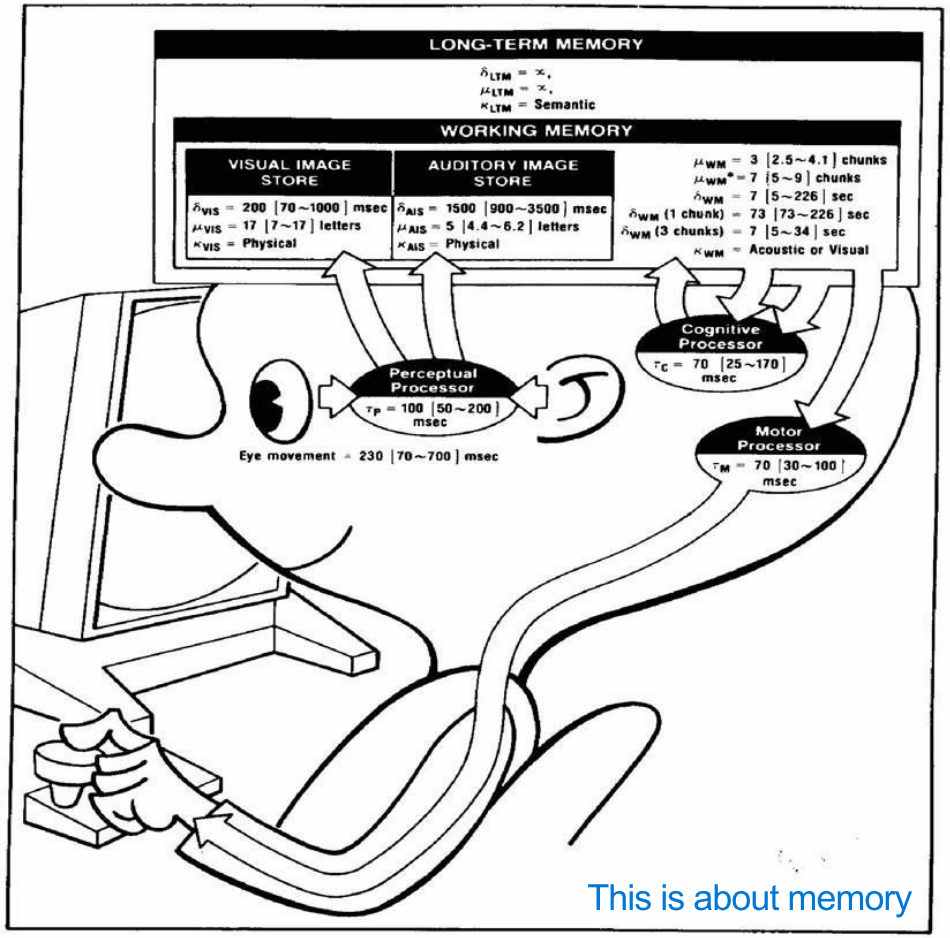
- Inconsistencies might be used (sparingly!) for drawing attention (e.g. use red for errors)

Consistent
delete/insert character
delete/insert word
delete/insert line
delete/insert paragraph

Human Processor Model (theory)

less is better (humans are not good in memory)

- Cognitive modelling method used to **calculate** how long it takes to perform a given task
  - prediction the system's performance (time to complete a task)
  - analogy between processing and storage areas of a computer with the perceptual, motor, cognitive and memory areas (working and long-term) of a person
- The calculations can be also used to determine the probability of a user remembering an item encountered during the task
- Underlies other usability techniques (GOMS, KLM, …)



Card, Stuart K.; Moran, Thomas P; and Newell, Allen. Human-computer interaction – Psychological aspects, Erlbaum Associates, 1983, ISBN: 9780898592436

Memory

- Working memory (short-term)
  - small capacity (7 ± 2 “chunks”)
    - +393475812632 vs. (+39) 347 581 2632
    - FGIHHJLMQ vs. FGI HHJ LMQ

broken in chuncks (more readable and memorable)

  - rapid access (~70ms) and decay (~200ms)
    - pass to long-term memory after a few seconds of continued storage
- Long-term memory
  - huge (unlimited, almost)
  - slower access time (~100ms) with litte decay

Fitts's Law

- Demonstration: https://fww.few.vu.nl/hci/interactive/fitts/

Based on mathematical calculations

- "The amount of time required for a person to move a *pointer* to a target area is a function of the distance to the target divided by the size of the target"
  - the longer the distance and the smaller the target's size, the longer it takes
  - created by psychologist Paul Fitts in 1954 examining the human motor system

- Widely used in HCI:
  - influenced the convention of making interactive buttons large (especially on finger-operated mobile devices)
  - the distance between a user's task/attention area and the task-related button should be kept as short as possible

Design Principles

Define how an interface is well designed?

The important aspects that we need to consider when creating a design.

The “What”

Design Principles

In the storyboard, you don't care about design principles (there is no interface)

- More practical than Theories but not a Guideline (so not so specific of how to do a spacific things). An example of Guideline is to put a description to images on HTML (tag)
- More fundamental, widely applicable, and enduring than Guidelines
- Fundamental principles (→ from Needfinding)
  - Determine user's skill levels
  - Identify the tasks
- 5 primary interaction styles (determine how humans and the system interact. Design principles can be applied across all main interaction styles)
- 8 golden rules of interface design (each rule is very big)
- Prevent errors
- Automation and human control



The main 5

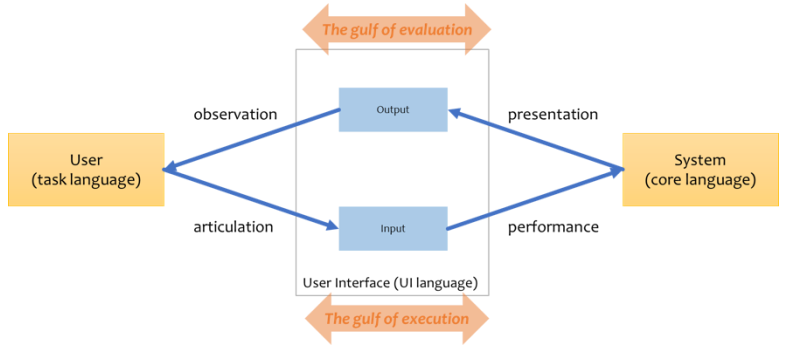
Interaction Styles

1. Direct manipulation (e.g., Drag & Drop)
2. Menu selection (e.g. in login/registration)
3. Form fill-in (e.g. terminal)
4. Command language
5. Natural language

Not to memorise

Advantages	Disadvantages
Direct manipulation	
Visually presents task concepts	May be hard to program
Allows easy learning	May require graphics display and pointing devices
Allows easy retention	
Allows errors to be avoided	
Encourages exploration	
Affords high subjective satisfaction	
Menu selection	
Shortens learning	Presents danger of many menus
Reduces keystrokes	May slow frequent users
Structures decision making	Consumes screen space
Permits use of dialog-management tools	Requires rapid display rate
Allows easy support of error handling	
Form fill-in	
Simplifies data entry	Consumes screen space
Requires modest training	
Gives convenient assistance	
Permits use of form-management tools	
Command language	
Flexible	Poor error handling
Appeals to "power" users	Requires substantial training and memorization
Supports user initiative	
Allows convenient creation of user-defined macros	
Natural language	
Relieves burden of learning syntax	Requires clarification dialog
	May not show context
	May require more keystrokes
	Unpredictable

Norman's Principles from Action Models



Per supportare l'utente nelle varie fasi dell'azione, un'interfaccia deve rispettare questi quattro requisiti fondamentali

Principles of good design

- State and the action alternatives should be visible (of the system)
- Should be a good conceptual model with a consistent system image (e.g. the human see the icon of a bin as to delete files or something: consistent with mental model of user)
- Interface should include good mappings that reveal the relationships between stages
- User should receive continuous feedback (to prevent errors) (not always pop-ups: are annoying if too much)

Quando questi principi non vengono rispettati, l'utente può riscontrare diverse difficoltà durante l'interazione

User failures can occur

- Users can form an inadequate goal
- Might not find the correct interface object because of an incomprehensible label or icon
- May not know how to specify or execute a desired action
- May receive inappropriate or misleading feedback



The 8 Golden Rules of Interface Design

- Strive for consistency
- Cater to universal usability
- Offer informative feedback
- Design dialogs to yield closure
- Prevent errors
- Permit easy reversal of actions
- Keep users in control
- Reduce short-term memory load

The 8 Golden Rules of Interface Design

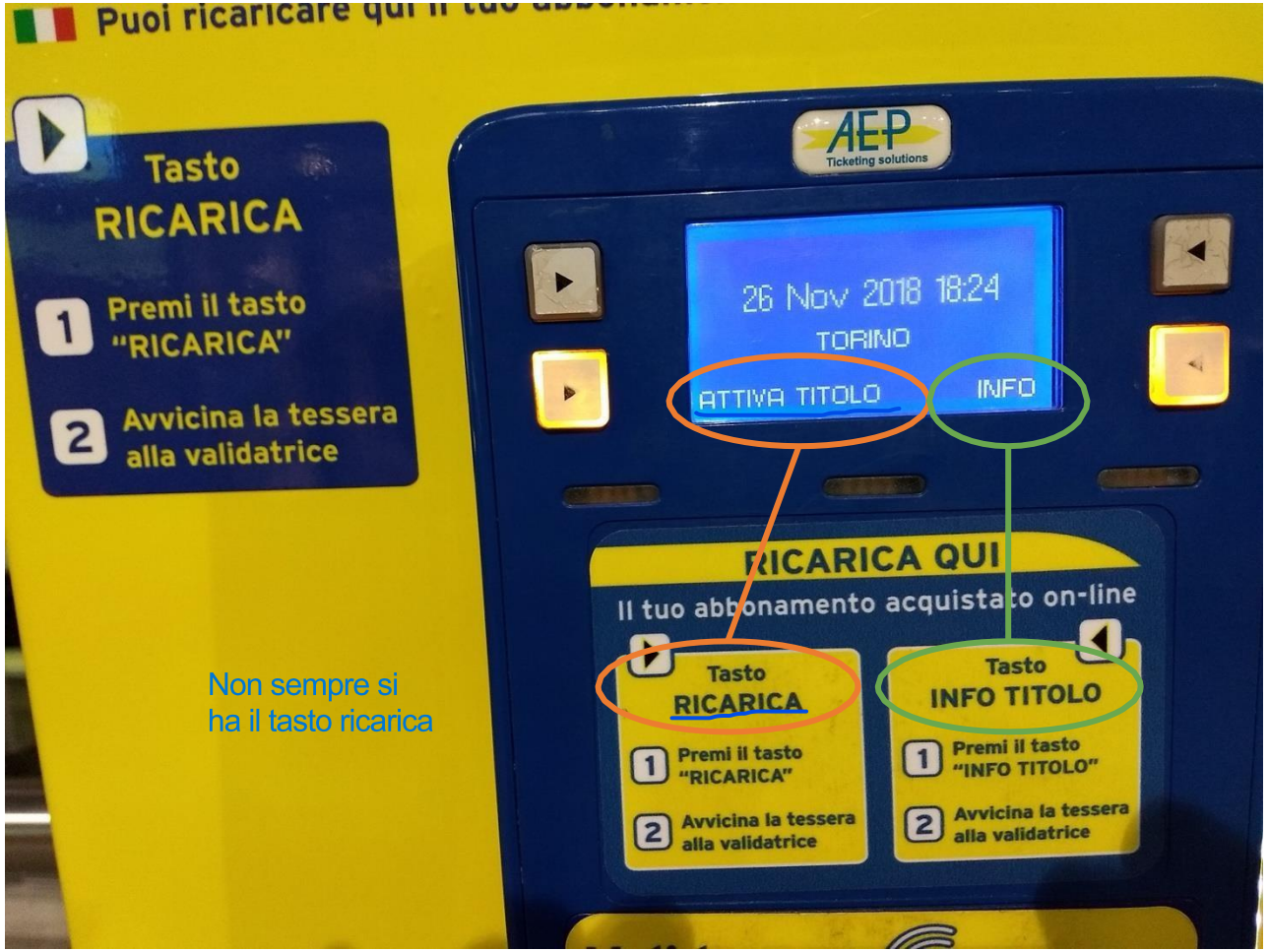
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- Similar situations should lead to similar sequences of actions (stessa sequenza porta a stesso risultato)
- Same terminology in prompts, menus, help

All related to interface

- Color, layout, capitalization, fonts, …
- Exceptions should be comprehensive and limited
  - E.g., delete, password echo

Internal Consistency



Consistency with mental models of the user

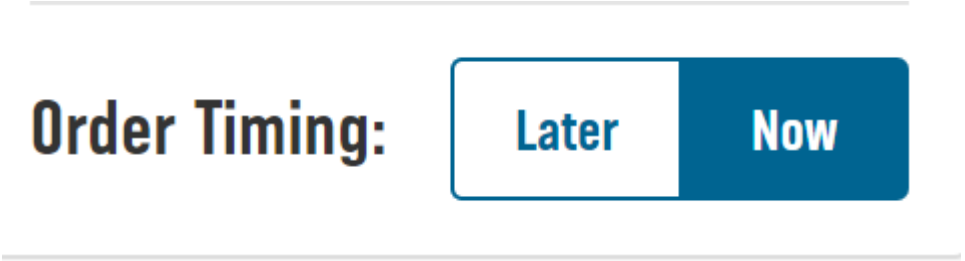






https://twitter.com/grmcall/status/1182586857814659073?s=20

Consistency of Interpretation



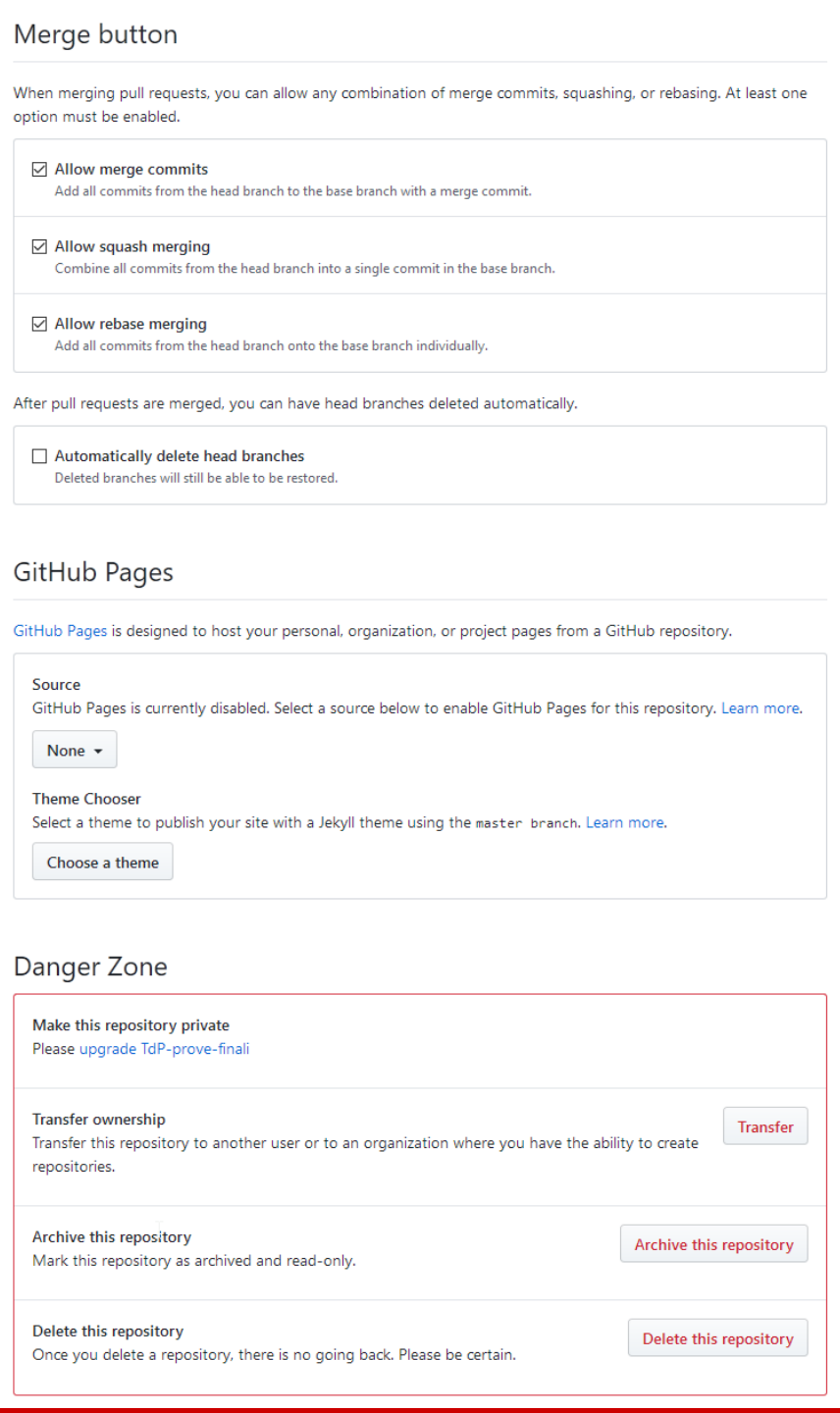
- Which one is the selected one?
  - Color codes are ambiguous
  - No further internal clues
  - No external clues
- Does it represent the current status?
- Does it represent the status that we want to achieve?

Perfect example

Inconsistency for Drawing Attention

The border color and button text color in the “danger zone” are deliberately different than the rest of the page

Not use only colors: keep in mind color issues (someone cannot distinguish some type of colors)





Inclusive design: different interfaces based on user capability/preferences
Universal design: create one interface that can be used by anyone (different from universal usability)

The 8 Golden Rules of Interface Design

- Strive for consistency
- **Cater to universal usability** (not so simple to implement)
- Offer informative feedback
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- Reduce short-term memory load

Ricorda che comunque devi avere un target user su cui basare il design principale

- Users with different needs: let the interface *adapt*, let content be *transformed* (not always so easy to adapt)

Different design for one to another

- Novices vs. experts. Young vs elderly. Web vs. mobile. Users with disabilities (→Accessibility)
- **Responsive** design
- International (and cultural) variations (e.g. language, color meaning)

The 8 Golden Rules of Interface Design

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- For ***every*** human action, there should be an interface feedback
- Frequent and minor actions: light feedback
- Infrequent and major actions: stronger feedback (e.g. pop-up)
- Visual presentation of objects helps showing the changes (e.g., dim, highlight, grey out, …)

Example

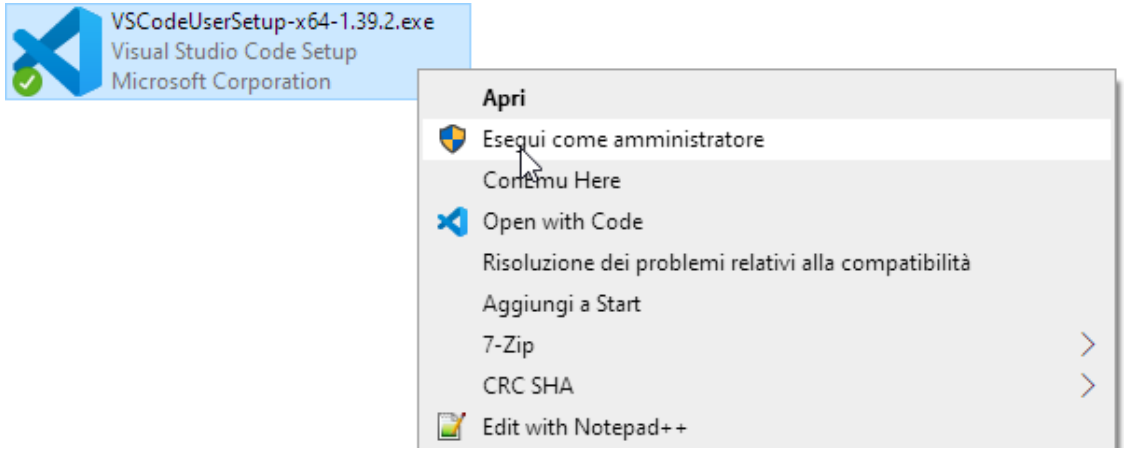
Bad example

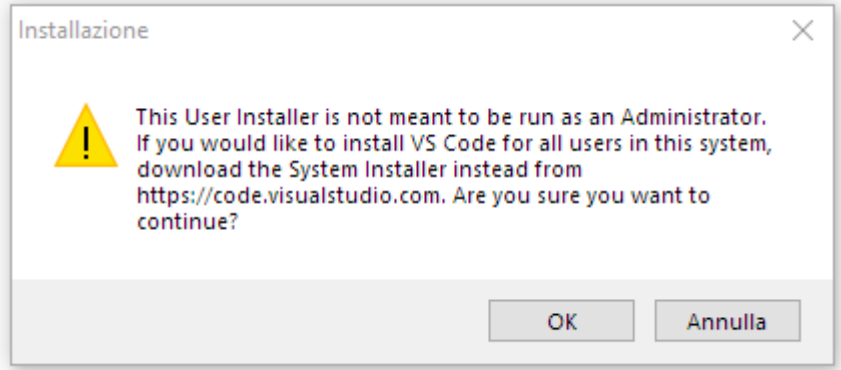


Which problem? What the user can do to solve?

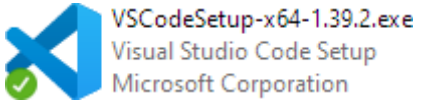
Example

Good example





Try to install VS Code for all users on a computer (install to Program Files rather than user's folders)



The 8 Golden Rules of Interface Design

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- Offer informative feedback
- **Design dialogs to yield closure**
- Prevent errors
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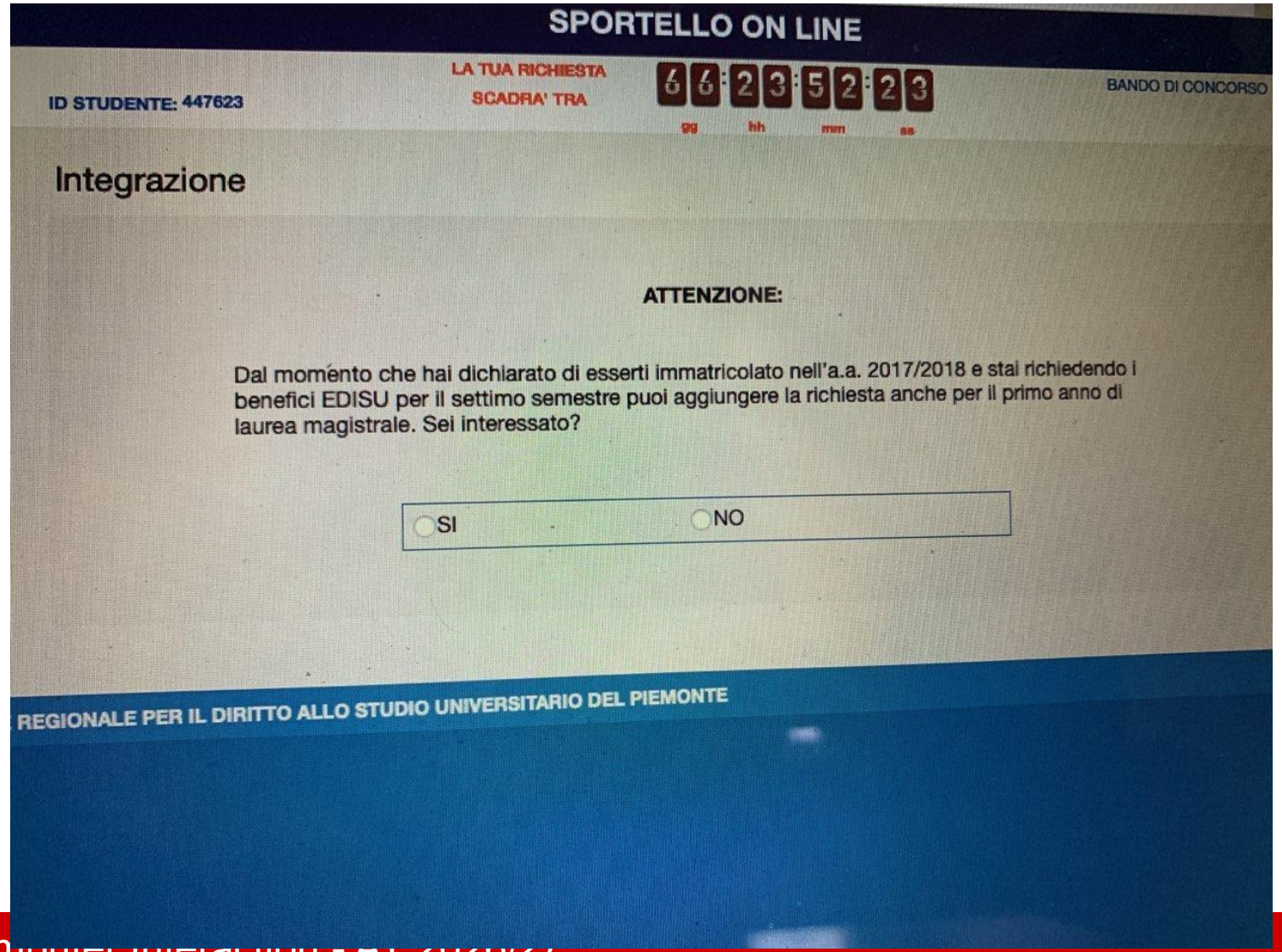
Related to Norman model

- Every sequence of actions should have
  - Beginning
  - Development
  - End
- Provide clear feedback at end
  - Satisfy users
  - 'Delete' current task from ~~their~~ of the user working memory, prepare for the next

Una volta terminata la sequenza, il sistema invia un segnale inequivocabile di successo. Questo permette all'utente di "cancellare" l'attività corrente dalla propria mente e prepararsi psicologicamente a quella successiva, senza il dubbio di aver dimenticato qualcosa.

Clear Dialog Sequence

Not clear about the sequence



- OK
- Submit
- Confirm
- Next
- …?

The 8 Golden Rules of Interface Design

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- **Prevent errors**
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Related to Norman model

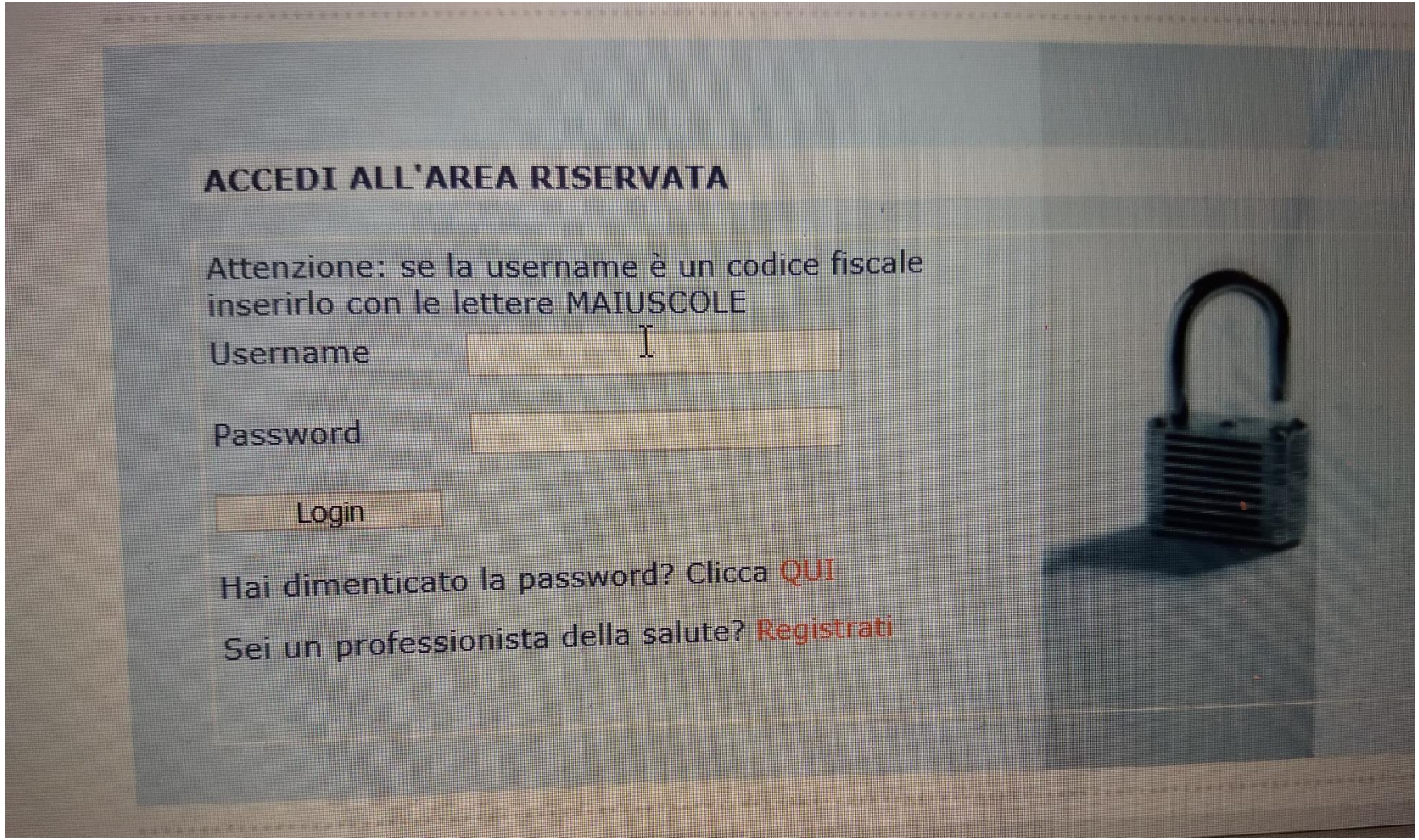
but still possible to do them

- Avoid the possibility of making errors
- Disable menu items, buttons, links, … that are not applicable
- Prevent entering illegal characters
- Offer simple, constructive and specific instructions for recovery
  - Repair only the faulty part
- Errors should not alter application state (or make it easy to restore)

Error Prevention

Bad example

Prevent errors by code



The 8 Golden Rules of Interface Design

- Strive for consistency
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- Prevent errors
- **Permit easy reversal of actions**
- Keep users in control
- Reduce short-term memory load

- Actions should be reversible (at the cost of extra development effort)
  - ~~Relieves~~ Reduces anxiety of user
  - Encourages exploration to user
- Different levels of reversibility
  - A single action
  - A data-entry task
  - A complete group of actions

The 8 Golden Rules of Interface Design

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- Design dialogs to yield closure
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- **Keep users in control**
- Reduce short-term memory load

The focus is on interface, not on user

- The interface should *always* respond to user actions
- Minimize the tedious and ~~lengthy~~ (very long) tasks
- Avoid surprises or changes in familiar behavior
- Provide undo/redo, cancel/confirm



Bad example: can put "no problem" and "have problems" as answer

Example

*Come docente, quali problemi hai avuto nello svolgimento degli esami?

Scegliere una o più delle seguenti opzioni

- [x] Non ho avuto problemi
- [x] Organizzazione dell'esame (poca chiarezza nella spiegazione delle modalità, sovrapposizione di date, procedure troppo confuse, deposito e consultazione documentazione complesso, ecc.)
- [ ] Dispongo di hardware/software inadeguato
- [ ] La connessione che uso è lenta/non continua
- [ ] Problemi ambientali (troppo rumore, confusione, scarsa possibilità di concentrazione)

The 8 Golden Rules of Interface Design

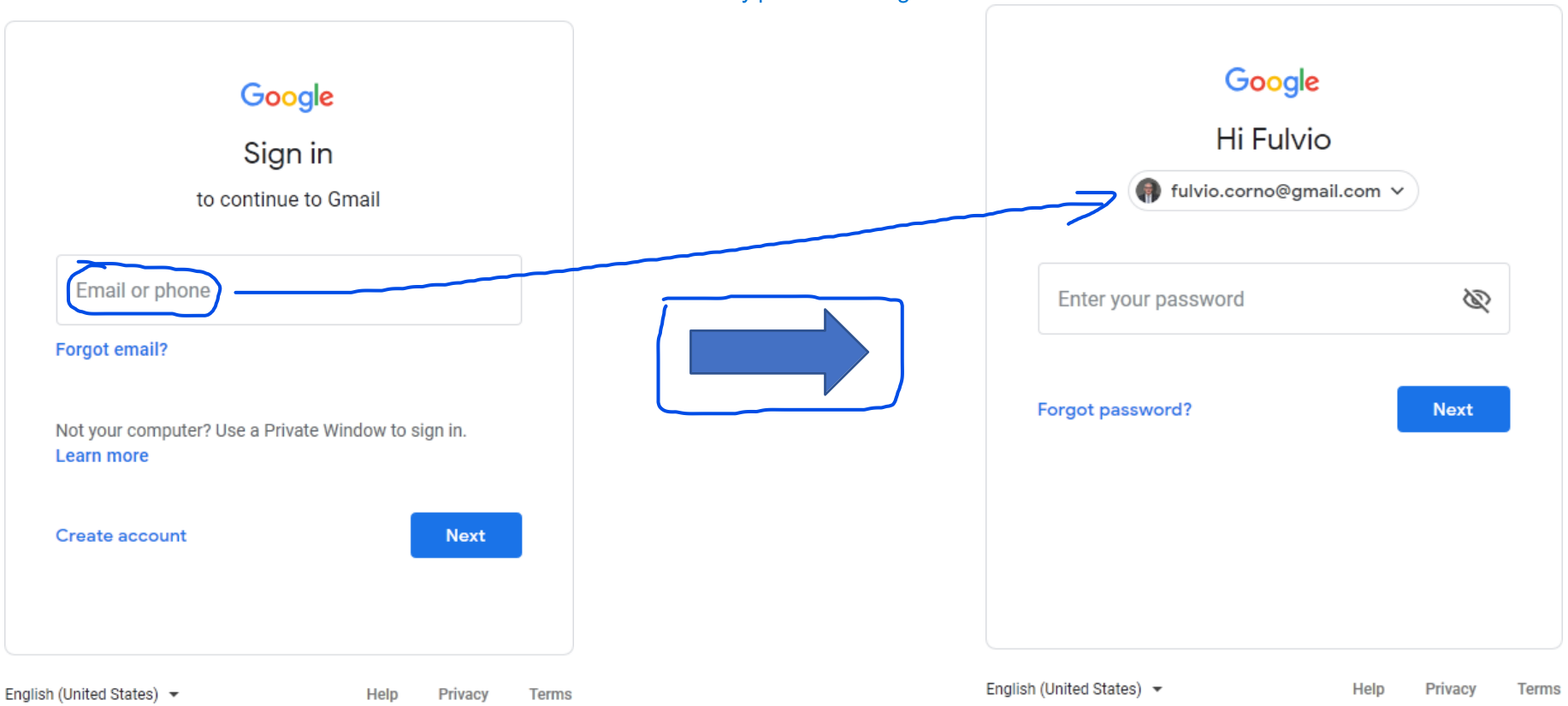
- Strive for consistency
- Cater to universal usability
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- Keep users in control
- **Reduce short-term memory load**

- Rule of thumb:
  - People can remember 7±2 chunks of information
- Information on a screen ~~should~~ that should be put not be needed (remembered) in the next screen
  try to divide information in different screens (but not too many screens)
- No entry of phone numbers (collect from addressbook), show website location, fit long forms in a single page…

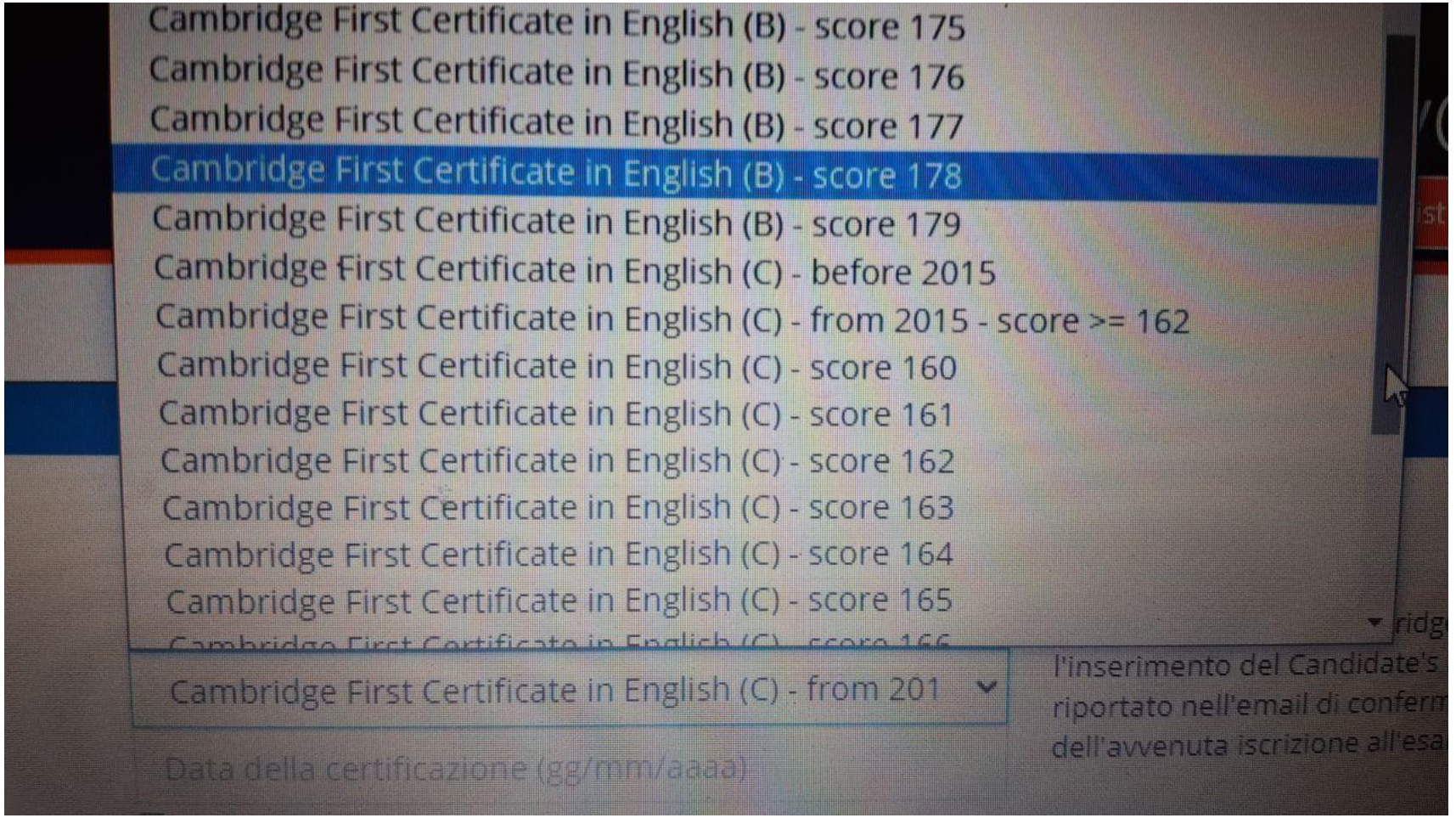
Discussion – An Exception?

Good idea in this case (not always): generally is better one screen so the user see what email typed

In this case, they used two screen to redirect to other domain if they put a different gmail mail



Exceptions...
sometimes entering is better than selecting





other principles by other researches

Design Principles by Benyon (I) (adapted from Norman, Nielsen and others)

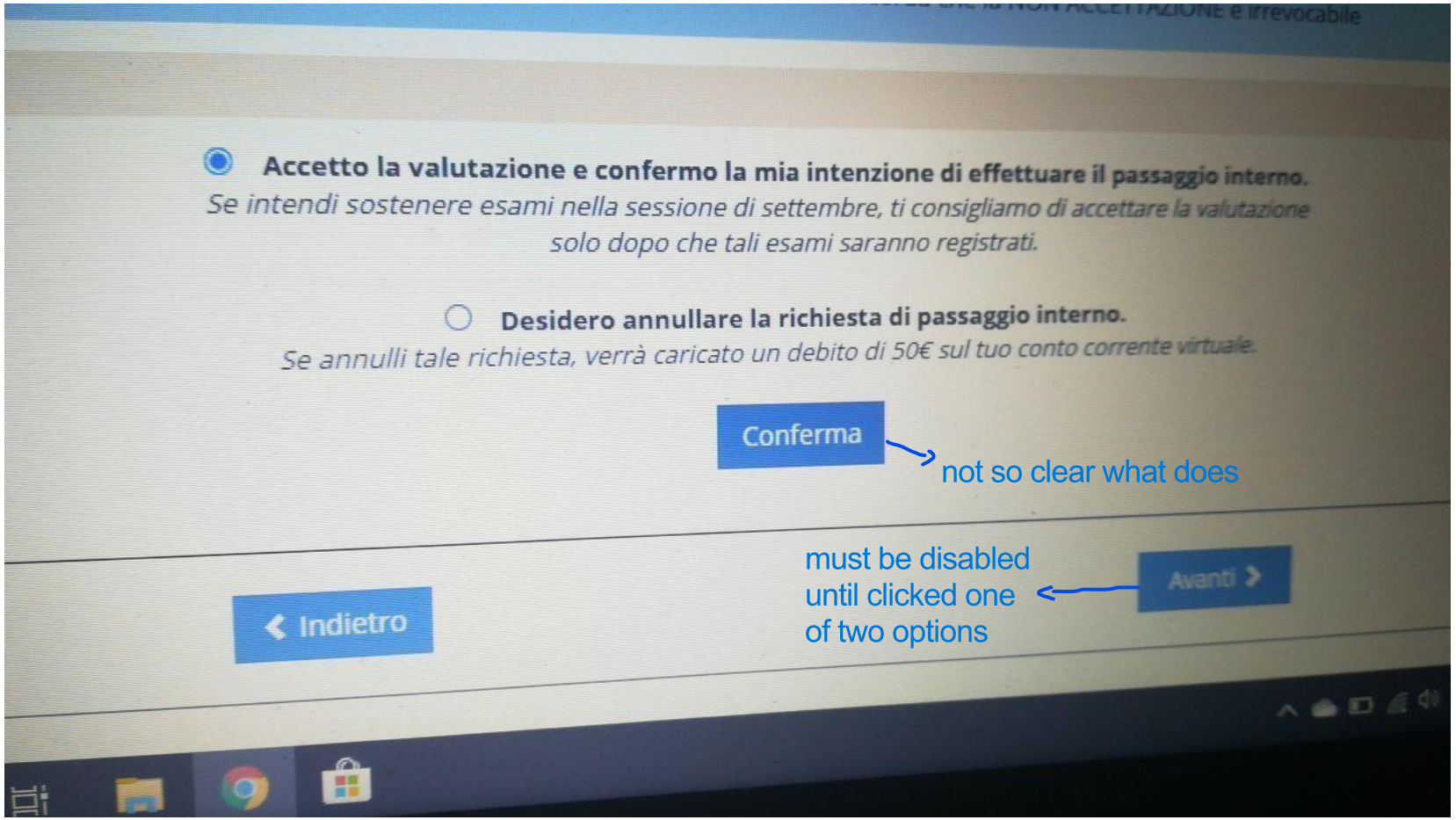
This is a new set of principles (adapted from the 8 golder rules)

- **Learnability** – helping people access, learn and remember the system
  - *Visibility* – ensure that things are visible, so users can see what functions are available and what the system is currently doing
  - *Consistency* (→above)
  - *Familiarity* – use language and symbols that the intended audience will be familiar with
  - *Affordance* – design things so it is clear what they are for (e.g., buttons should be pushed). Maps the (perceived) properties of the objects with how they can be used

Design Principles by Benyon(II) (adapted from Norman, Nielsen and others)

- **Effectiveness** – giving users the sense of being in control, knowing what to do and how to do it
  - *Navigation* – support people in moving around the different sections: maps, directional signs, information signs
  - *Control* – who is in control for the next interaction? Clear and logical mapping between controls and their effect. Relationships with the "side effects" in the real world
  - *Feedback* (→feedback above)

Example: Navigation and Control?



Design Principles by Benyon(III) (adapted from Norman, Nielsen and others)

- **Safety and Security**
  - *Recovery* (→error recovery)
  - *Constraints* (→prevent errors)
- **Accommodation** – offer an interaction way that suits the users
  - *Flexibility* (→universal usability)
  - *Style* – stylish, attractive, nice-looking
  - *Conviviality* – polite, friendly, pleasant. No abrupt interruptions

with user

Norman's Seven Principles for Transforming Difficult Tasks into Simple Ones

Skipped

- Use both knowledge in the world and knowledge in the head
- Simplify the structure of tasks
- Make things visible
- Get the mappings right
- Exploit the power of constraints, both natural and artificial
- Design for error
- When all else fails, standardize

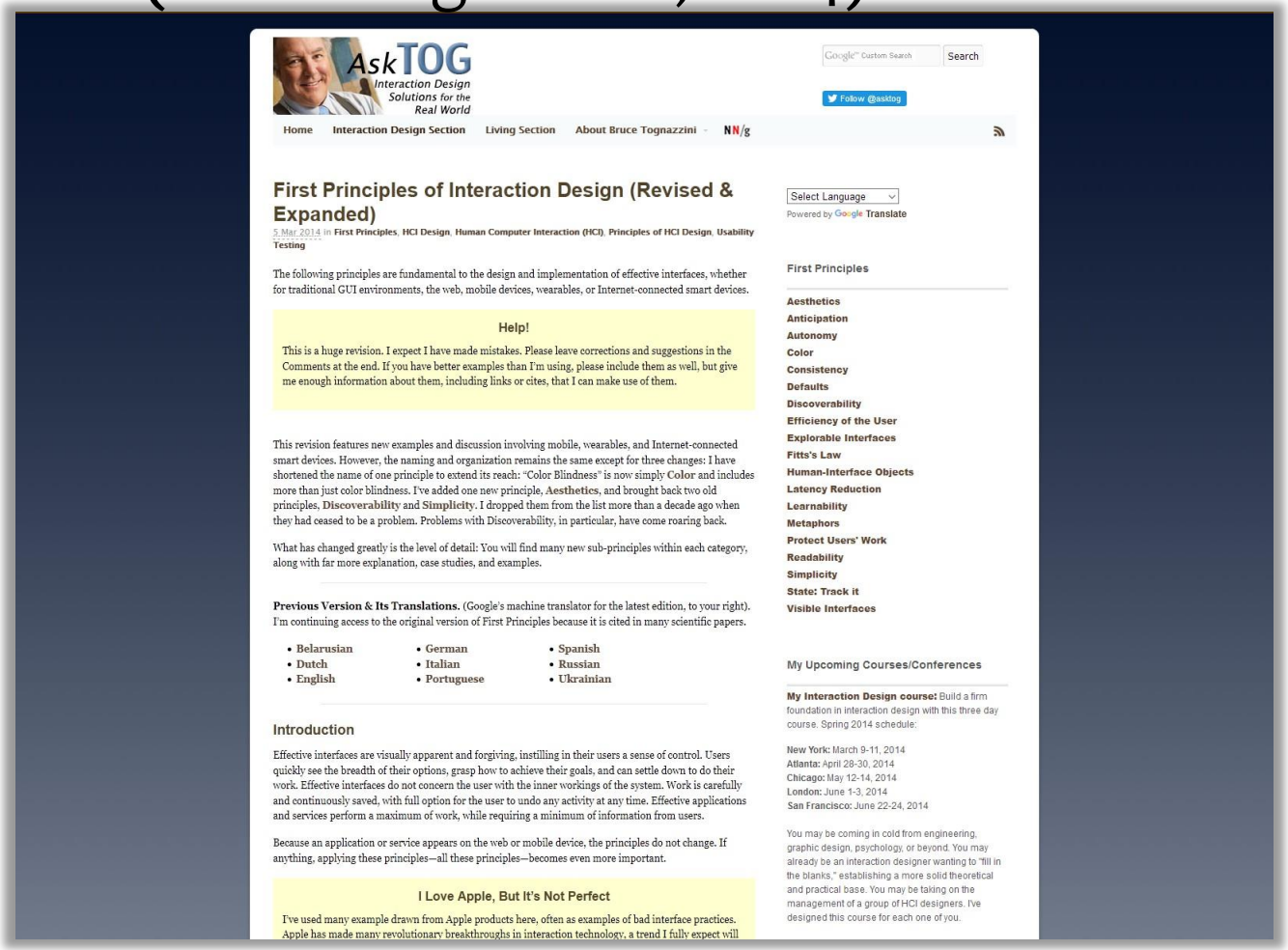
D. Norman, The Design of Everyday Things

First Principles of Interaction Design

(Bruce Tognazzini, 2014)

https://asktog.com/atc/principles-of-interaction-design/

Skipped



Aesthetics
Anticipation
Autonomy
Color
Consistency
Defaults
Discoverability
Efficiency of the User
Explorable Interfaces
Fitts's Law
Human-Interface Objects
Latency Reduction
Learnability
Metaphors
Protect Users' Work
Readability
Simplicity
State: Track it
Visible Interfaces

References and Acknowledgments

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- David Benyon: Designing Interactive Systems, Pearson, 2014
  - Section 4.5: Design Principles
- COGS120/CSE170: Human-Computer Interaction Design, videos by Scott Klemmer,
- https://www.youtube.com/playlist?list=PLLssT5z_DsK_nusHL_Mjt87THSTlgrsyJ
- Fitts' Law: https://www.interaction-design.org/literature/topics/fitts-law

- These slides are a modified version of the ones of prof. Luigi De Russis, PoliTo. A big thanks to Luigi De Russis (PoliTo) for his work!